1. Vertical Motion

1.1. Introduction

- **Weight:** directly downwards
- **Normal contact force:** perpendicular to place of contact

1.2. Common Results of Vertical Motion

Finding time taken to reach maximum height by a projectile travelling in vertical motion:

$$v = u + at$$

- Let $v = 0$ and find t
- The time taken to go up and come back to original position would be double of this t

Finding maximum height above a launch point use:

- $v^2 = u^2 + 2as$
- Let $v = 0$ and find s

Finding time interval for which a particle is above a given height:

- Let the height be H and use
- $s = ut + \frac{1}{2}at^2$
- Let $s = H$
- There will be a quadratic equation in t
- Solve and find the difference between the 2 t's to find the time interval

{S04-P04} Question 7:

Particle P_1 projected vertically upwards, from the horizontal ground, with speed $30ms^{-1}$. At same instant P_2 projected vertically upwards from tower height $25m$, with speed $10ms^{-1}$

1. Find the time for which P_1 is higher than the top of the tower
2. Find the velocities of the particles at the instant when they are the same height
3. Find the time for which P_1 is higher than P_2 and moving upwards

Solution

Part (i)

Substitute given values into displacement equation:

$$25 = (30)\,t - \frac{1}{2}(10)\,t^2$$

$$5t^2 - 30t + 25 = 0$$

Solve quadratic for t

$$t = 1s \text{ or } 5s$$

P_1 reaches tower at $t = 1$ then passes it again when coming down at $t = 5s$

Therefore, time above tower $= 5 - 1 = 4$ seconds

Part (ii)

Displacement of P_1 is S_1, and of P_2 is S_2 & relationship:

$$S_1 = 25 + S_2$$

Create equations for S_1 and S_2

$$S_1 = 30t + \frac{1}{2}(-10)\,t^2 \qquad S_2 = 10t + \frac{1}{2}(-10)\,t^2$$

Substitute back into the initial equation

$$30t + \frac{1}{2}(-10)\,t^2 = 25 + 10t + \frac{1}{2}(-10)\,t^2$$

Simple cancelling

$$t = 1.25s$$

Find velocities

$$v = u + at$$

$$V_1 = 30 - 10\,(1.25) = 17.5ms^{-1}$$

$$V_2 = 10 - 10\,(1.25) = -2.5ms^{-1}$$

Part (iii)

We know when P_1 and P_2 at the same height $t = 1.25s$. Find the time taken to reach max height for P_1

$$v = u + at$$

V is 0 at max height

$0 = 30 - 10t \quad t = 3s$

Time for P_1 above P_2 $= 3 - 1.25 = 1.75$ seconds

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